

Partial Differentiation – Solutions

LLE – Mathematics and Statistics Skills

Differentiation

1. (a) For $y = 5x^3 + 10x - 8$:

$$\frac{dy}{dx} = 15x^2 + 10$$

- (b) For $y = \frac{10}{x^2} - 5\sqrt{x}$:

Rewrite as $y = 10x^{-2} - 5x^{1/2}$.

$$\frac{dy}{dx} = -20x^{-3} - \frac{5}{2}x^{-1/2}$$

- (c) For $y = \ln(8x)$:

Rewrite as $y = \ln(8) + \ln(x)$.

$$\frac{dy}{dx} = \frac{1}{x}$$

- (d) For $y = e^{5x} + 3e^{x^2}$:

$$\frac{dy}{dx} = 5e^{5x} + 6xe^{x^2}$$

Partial Derivatives

2. (a) For $z = x^2 - y^2$:

$$\frac{\partial z}{\partial x} = 2x$$

$$\frac{\partial^2 z}{\partial x^2} = 2$$

$$\frac{\partial^2 z}{\partial y \partial x} = 0$$

$$\frac{\partial z}{\partial y} = -2y$$

$$\frac{\partial^2 z}{\partial y^2} = -2$$

$$\frac{\partial^2 z}{\partial x \partial y} = 0$$

(b) For $z = 5x + 3x^2 - 2y^3$:

$$\frac{\partial z}{\partial x} = 5 + 6x$$

$$\frac{\partial^2 z}{\partial x^2} = 6$$

$$\frac{\partial^2 z}{\partial y \partial x} = 0$$

$$\frac{\partial z}{\partial y} = -6y^2$$

$$\frac{\partial^2 z}{\partial y^2} = -12y$$

$$\frac{\partial^2 z}{\partial x \partial y} = 0$$

(c) For $z = xy$:

$$\frac{\partial z}{\partial x} = y$$

$$\frac{\partial^2 z}{\partial x^2} = 0$$

$$\frac{\partial^2 z}{\partial y \partial x} = 1$$

$$\frac{\partial z}{\partial y} = x$$

$$\frac{\partial^2 z}{\partial y^2} = 0$$

$$\frac{\partial^2 z}{\partial x \partial y} = 1$$

(d) For $z = x^2y^3 + 8x - 4$:

$$\frac{\partial z}{\partial x} = 2xy^3 + 8$$

$$\frac{\partial^2 z}{\partial x^2} = 2y^3$$

$$\frac{\partial^2 z}{\partial y \partial x} = 6xy^2$$

$$\frac{\partial z}{\partial y} = 3x^2y^2$$

$$\frac{\partial^2 z}{\partial y^2} = 6x^2y$$

$$\frac{\partial^2 z}{\partial x \partial y} = 6xy^2$$

(e) For $z = e^{x^2-y^2}$:

First-Order Derivatives (using the chain rule):

$$\begin{aligned}\frac{\partial z}{\partial x} &= e^{x^2-y^2} \cdot (2x) = 2xe^{x^2-y^2} \\ \frac{\partial z}{\partial y} &= e^{x^2-y^2} \cdot (-2y) = -2ye^{x^2-y^2}\end{aligned}$$

Second-Order Derivatives (using the product rule):

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x}(2xe^{x^2-y^2}) = (2)(e^{x^2-y^2}) + (2x)(2xe^{x^2-y^2}) \\ &= 2e^{x^2-y^2}(1 + 2x^2) \\ \frac{\partial^2 z}{\partial y^2} &= \frac{\partial}{\partial y}(-2ye^{x^2-y^2}) = (-2)(e^{x^2-y^2}) + (-2y)(-2ye^{x^2-y^2}) \\ &= 2e^{x^2-y^2}(-1 + 2y^2)\end{aligned}$$

Cross-Partial Derivatives:

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}(2xe^{x^2-y^2}) = -4xye^{x^2-y^2}$$

Implicit Differentiation

3. (a) For $x^2 + 4y = 0$:

$$2x + 4\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{2}$$

(b) For $x^2 - y^2 = 0$:

$$2x - 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{x}{y}$$

(c) For $4xy + y^2 = 0$:

Using the product rule on $4xy$:

$$(4y + 4x \frac{dy}{dx}) + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{2y}{2x + y}$$

Total Differential

4. (a) For $z = x^2 + 4y$: $dz = 2x dx + 4 dy$.
 (b) For $z = x^2 - y^2$: $dz = 2x dx - 2y dy$.
 (c) For $z = 4xy + y^2$: $dz = 4y dx + (4x + 2y) dy$.
 (d) For $z = \frac{4x^2}{y^3}$: $dz = \frac{8x}{y^3} dx - \frac{12x^2}{y^4} dy$.
5. *Solution:* The total differential is $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$. If $f(x, y) = 0$, then $dz = 0$.

$$0 = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$\frac{dy}{dx} = -\frac{\partial z / \partial x}{\partial z / \partial y}$$

- (a) For $z = x^2 + 4y$: $\frac{dy}{dx} = -\frac{2x}{4} = -\frac{x}{2}$. This matches the result from the 'Implicit Differentiation' section.
- (b) For $z = x^2 - y^2$: $\frac{dy}{dx} = -\frac{2x}{-2y} = \frac{x}{y}$. This matches.
- (c) For $z = 4xy + y^2$: $\frac{dy}{dx} = -\frac{4y}{4x+2y} = -\frac{2y}{2x+y}$. This matches.
- (d) For $z = \frac{4x^2}{y^3}$: $\frac{dy}{dx} = -\frac{8x/y^3}{-12x^2/y^4} = \frac{2y}{3x}$.